

EC3218 — Portfolio Analysis

Summer 2024 Examination · Paper 1 · Full Marking Scheme

Module	EC3218 — Portfolio Analysis
Internal Examiner	Professor Niall O'Sullivan
Institution	University College Cork
Session	Summer 2024, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal marks.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

A different sitting, a different emphasis. This paper draws on a more specialised, research-level segment of the module than the Summer 2025 EC3218 sitting — market timing, conditional beta models, volatility timing, and performance persistence testing are all active academic research areas (several of them associated with Professor O'Sullivan's own published work). The Fama-French content in Q. A1(a) is the same underlying model as the Summer 2025 scheme, but this question asks you to describe it fully in its own right rather than compare it to CAPM.

Source basis. Treynor-Mazuy (1966) is drawn directly from the module's own lecture material. The conditional beta model (Ferson and Schadt, 1996), the volatility timing literature (Busse, 1999), the momentum crash literature (Daniel and Moskowitz, 2016), and the performance persistence testing literature (Carhart, 1997; Hendricks, Patel and Zweig, 1993; Malkiel, 1995) were not set out in the uploaded course material and were verified against the original published papers —

Section A · Question 1

The Fama-French model, momentum trading, and conditional beta — 100 marks

A1(a)

[40 marks]

Fully describe the Fama and French three-factor model of portfolio performance.

A1(b)

[30 marks]

In portfolio management, explain what is meant by a momentum trading strategy. Does momentum in stock returns represent risk to a momentum trader? Explain your answer.

A1(d)

[30 marks]

Derive and interpret a conditional beta model of fund performance.

FULL MODEL ANSWER

The model: the Fama-French three-factor model explains a portfolio's returns using three factors rather than CAPM's single beta:

Model: $R_p - R_f = \alpha_p + \beta_1(R_m - R_f) + \beta_2 \cdot \text{SMB} + \beta_3 \cdot \text{HML} + \varepsilon_p$

- **Market factor, ($R_m - R_f$):** CAPM's own market risk premium, with sensitivity β_1 — identical in spirit to CAPM's beta.
- **Small Minus Big (SMB):** the excess return of a portfolio of small-cap stocks over a portfolio of large-cap stocks — constructed empirically each month as the return difference between the smallest 50% and largest 50% of firms by market capitalisation.
- **High Minus Low (HML):** the excess return of a portfolio of high book-to-market ("value") stocks over low book-to-market ("growth") stocks — constructed as the return difference between the top 30% and bottom 30% of firms by book-to-market ratio.

Using the model to evaluate portfolio performance: in a performance-evaluation context, the model is estimated as a time-series regression of a fund's excess returns on the three factors. The regression intercept, α_p ("**three-factor alpha**"), is the key output: it measures the fund's average

return *after* stripping out the compensation the fund earned simply for its exposure to all three systematic risk sources (market, size, value) — a genuinely more demanding benchmark than the single-factor Jensen's alpha (see the EC3218 Winter 2025 scheme, Q. A3(c)), since a fund that appears skilled under CAPM (positive market-only alpha) may simply be earning the well-documented small-firm or value premium, and show *no* abnormal three-factor alpha once SMB and HML exposure are properly accounted for.

The empirical findings that motivate using it for performance

evaluation: Fama and French, using US stock data from 1963-1991, found both SMB and HML carry statistically and economically significant positive premiums, and that the 3-factor model convincingly explains variation in returns that CAPM alone could not. Because a large share of what looks like manager "skill" under a simple CAPM benchmark can be reclassified as compensation for tilting a portfolio toward small and value stocks, the 3-factor model became the standard tool for separating genuine security-selection skill from mechanical factor exposure.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly states the full regression equation with all three factors AND correctly explains how SMB and HML are empirically constructed AND correctly explains the model's use in performance evaluation via the three-factor alpha, distinguishing it from single-factor (CAPM) alpha AND the empirical motivation (significant SMB/HML premiums) is given.
21-30	21-30	Correctly states the model and factor construction, but the performance-evaluation application (three-factor alpha vs. CAPM alpha) is not clearly explained.
11-20	11-20	Only the market factor and one of SMB/HML correctly described.
0-10	0-10	No credible description of the model given.

What a momentum trading strategy is: a momentum strategy exploits the well-documented tendency of past "winner" stocks (those with high returns over the past several months to a year) to continue outperforming, and past "loser" stocks to continue underperforming, over the following several months (Jegadeesh and Titman, 1993). The strategy itself is simple: go long a portfolio of recent winners and short a portfolio of recent losers, profiting from the continuation of the existing trend rather than betting on any reversal.

Does momentum in stock returns represent risk to a momentum trader? Yes — a specific, well-documented, and severe form of risk known as "momentum crash risk." Momentum strategies have historically delivered strong average returns and high Sharpe ratios, but Daniel and Moskowitz (2016) show that momentum returns are **negatively skewed**: the strategy experiences infrequent, but severe and persistent, strings of losses. Crucially, these crashes are not random — they occur specifically in "panic states," following market declines and periods of high market volatility, and are **contemporaneous with sharp market rebounds**. In a rebound following a downturn, past "loser" stocks (which a momentum strategy is short) tend to bounce back especially sharply, inflicting outsized losses on the short leg of the strategy at exactly the moment the long leg (past winners) is not compensating for it.

Why this matters for the momentum trader specifically: this crash risk is a real, structural feature of the strategy itself, not a risk that can be diversified away by holding more stocks within the winner/loser portfolios — it is a

systematic, market-timing-related risk tied to the state of the broader market (recent declines and elevated volatility specifically predict momentum crash risk). This is precisely why more recent research (e.g. Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016) has focused on **dynamically scaling** momentum exposure — reducing position size when momentum's own recent volatility signals an elevated crash risk — rather than treating momentum as a purely diversifiable anomaly with no risk of its own.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly explains the momentum strategy (long recent winners, short recent losers) AND correctly concludes momentum represents a real risk to the trader AND correctly explains the crash-risk mechanism (negative skewness, panic-state timing, losses concentrated around market rebounds following declines).
15-22	15-22	Correct strategy description and correct "yes, it's risky" conclusion, but the specific crash-risk mechanism is only asserted, not explained.
8-14	8-14	Correct strategy description, but concludes momentum carries no risk to the trader beyond ordinary market risk.
0-7	0-7	No credible description of the momentum strategy given.

EXAM TECHNIQUE

The conditional beta model wasn't set out in the uploaded course material; the derivation below follows the standard treatment (Ferson and Schadt, 1996) — the seminal, widely-used framework for this type of model.

The problem the model solves: a standard, unconditional performance regression (Jensen's alpha, or Treynor-Mazuy — see Q. A2(a)) implicitly assumes a fund's beta is **constant** over the sample period. In reality, a manager's true portfolio beta may shift predictably over time simply because it moves mechanically with widely available public information (e.g. the general level of interest rates, the dividend yield, the slope of the yield curve) — for instance, if a manager's typical holdings naturally become more market-sensitive when rates are low. If this predictable beta variation isn't accounted for, a standard market-timing test (see Q. A2(a)) can **mistakenly attribute this purely mechanical beta variation to genuine manager timing skill** — a serious source of bias.

Deriving the model: start from the assumption that the fund's true (conditional) beta is a linear function of a vector of lagged, publicly-observable instruments Z_{t-1} (e.g. the T-bill yield, the dividend yield, the term spread, the default spread):

Step 1 — conditional beta: $\beta_{p,t} = b_{0p} + b_{1p}' Z_{t-1}$

Step 2 — substitute into the standard performance regression: $R_{p,t} - R_{f,t} = \alpha_p + \beta_{p,t} (R_{m,t} - R_{f,t}) + \varepsilon_{p,t}$

Step 3 — expand: $R_{p,t} - R_{f,t} = \alpha_p + b_{0p}(R_{m,t} - R_{f,t}) + b_{1p}' [Z_{t-1}(R_{m,t} - R_{f,t})] + \varepsilon_{p,t}$

Interpreting the model: the resulting regression is estimable directly by OLS — it looks like a standard multiple regression, but with an extra set of explanatory variables formed by **interacting** each lagged public instrument with the market excess return. b_{0p} is the fund's **average conditional beta**

(its beta when the instruments are at their average/zero-deviation level); b_{1p} captures how much the fund's beta moves in response to changes in each public information variable — a manager whose beta genuinely does track public information (rather than exercising discretionary skill) will show significant b_{1p} coefficients. α_p in this conditional specification is the fund's **conditional alpha** — performance measured *net of* any return attributable to the fund's beta mechanically tracking public information, leaving a cleaner measure of genuine, information-independent skill than the unconditional alpha would give.

Why this matters in practice — the "beta puzzle": Ferson and Schadt (1996) found that failing to condition on public information can materially bias performance and timing measures — indeed, subsequent research has documented a puzzling tendency for the average mutual fund's conditional beta to rise when the market risk premium falls (the opposite of what skilled timing would predict), underscoring why conditioning on public information, rather than assuming a constant beta, is now standard practice in rigorous fund performance evaluation.

Mark Allocation — Part (d) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies the problem (mechanical beta variation being mistaken for timing skill) AND correctly derives the conditional beta specification and its substitution into the performance regression AND correctly interprets b_0 , b_1 , and the conditional alpha.
15-22	15-22	Model correctly stated and substituted, but the motivating problem or the interpretation of the coefficients is incomplete.
8-14	8-14	Vague reference to "beta can change over time" without the specific linear-in-instruments derivation.
0-7	0-7	No credible derivation given.

Question A1 Total: 100%

Section A · Question 2

The Treynor-Mazuy market timing model, and a model of volatility timing — 100 marks

A2(a)

[50 marks]

Fully describe, with the aid of a diagram, the Treynor-Mazuy model of market timing.

Discuss any one model of volatility timing.

FULL MODEL ANSWER

What market timing is: in its pure form, market timing involves a manager shifting funds between a risky market-index portfolio and a safe asset (e.g. T-bills) in anticipation of market movements — increasing market exposure ahead of an anticipated market rise, and reducing it ahead of an anticipated decline.

The model: Treynor and Mazuy (1966) proposed testing for timing ability by adding a squared market-excess-return term to the standard performance regression:

$$\text{Model: } R_p - R_f = a + b(R_m - R_f) + c(R_m - R_f)^2 + e$$

Why the squared term captures timing ability: a manager with genuine timing ability increases market exposure (raises the portfolio's effective beta) precisely when market returns are about to be high, and decreases exposure when market returns are about to be low. This means the relationship between the portfolio's excess return and the market's excess return is no longer a simple straight line (as it would be for a manager holding a constant beta) — it becomes **convex**, curving upward at both large positive and large negative market excess returns, since a good timer earns more than a constant-beta manager on the market's best days (having raised exposure into them) and loses less on its worst days (having reduced exposure ahead of them).

Steeper slope when
market returns are high:
manager raises exposure
ahead of market upswings

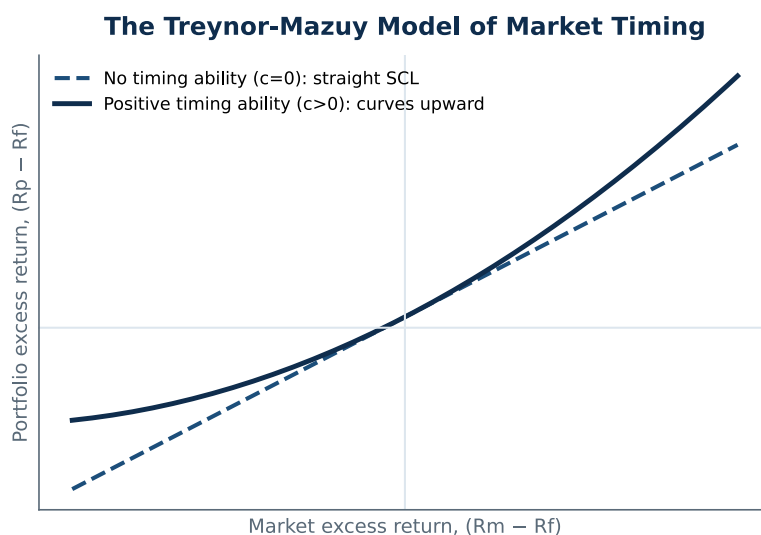


Fig. 1 — A manager with no timing ability ($c=0$) traces a straight security characteristic line. A manager with genuine timing ability ($c>0$) traces a curve that is steeper at high market returns — the hallmark of raising exposure ahead of market upswings.

Interpreting the key coefficient, c: c is the key coefficient in the model. If c is **positive and statistically significant**, this indicates genuine market timing ability — the manager's effective beta is systematically higher when subsequent market returns are higher, exactly the convex pattern a successful timer should produce. If c is zero or negative, there is no evidence of (successful) timing — a negative c would actually indicate the manager's exposure moves in the *wrong* direction relative to subsequent market moves.

EXAM TECHNIQUE

A well-known practical wrinkle (also covered in the module's own material): if a manager successfully times the market by genuinely varying beta, but the tester instead estimates a single, unconditional Sharpe ratio over the full sample, the resulting portfolio return series will look artificially more volatile than a constant-beta portfolio with the same average beta — this can make the Sharpe ratio understate a good timer's true risk-adjusted performance, which is part of why timing-specific models like Treynor-Mazuy (rather than the plain Sharpe ratio) are needed to properly credit timing skill.

Mark Allocation — Part (a) (50 marks)

Tier	Marks	What separates this tier
38-50	38-50	Correctly defines market timing AND correctly states the Treynor-Mazuy regression equation AND correctly explains why the squared term captures timing ability (convexity from raising exposure ahead of market upswings) AND the diagram correctly shows the curved vs. straight SCL AND correctly interprets the sign/significance of c .
25-37	25-37	Model and diagram correctly given, but the economic reasoning for why convexity indicates timing ability is only partially explained.
13-24	13-24	Model equation stated correctly, but the diagram or the interpretation of c is missing or incorrect.
0-12	0-12	No credible description of the model given.

EXAM TECHNIQUE

Volatility timing wasn't set out in the uploaded course material; the discussion below follows the standard treatment (Busse, 1999) — the seminal paper in this specific literature.

What volatility timing is: distinct from *return* timing (Q. A2(a)), **volatility timing** asks whether a manager adjusts market exposure in response to changes in expected market *volatility* specifically — reducing exposure ahead of periods of anticipated high volatility (to protect the portfolio from larger swings), and increasing exposure ahead of anticipated low-volatility periods.

The Busse (1999) model: Busse's key methodological insight was that volatility timing is very difficult to detect using standard **monthly** mutual fund return data, since monthly returns average away most of the day-to-day variation in volatility a manager might actually be responding to. Busse instead used **daily** mutual fund returns, testing whether a fund's beta (or market exposure) responds to changes in conditional market volatility, broadly of the form:

Model (illustrative form): $R_{p,t} - R_{f,t} = a + b(R_{m,t} - R_{f,t}) + c \cdot \text{var}_{m,t} \cdot (R_{m,t} - R_{f,t}) + e$

Interpretation: a **negative and significant c** indicates successful volatility timing — the manager's effective market exposure falls (the sensitivity of portfolio returns to market returns weakens) exactly when conditional market volatility is high, consistent with a manager deliberately reducing risk ahead of turbulent periods.

The findings: using daily data, Busse found that mutual fund managers do, on average, exhibit successful volatility timing — reducing market exposure ahead of high-volatility periods — and that this ability translates into **higher risk-adjusted returns (higher Sharpe ratios)** for the funds that exhibit it. Notably, Busse found this sensitivity to market volatility was considerably stronger among **surviving** funds than funds that were later shut down or merged away, suggesting volatility timing ability is genuinely linked to a fund's long-run success.

Why this matters alongside return timing: prior research using monthly data had generally found **little to no evidence** of successful return timing (Q. A2(a)) among mutual fund managers — Busse's contribution was to show that even where managers show no ability to time market *direction*, many still show genuine skill in timing market *risk*, a distinct and previously under-examined dimension of manager skill that monthly-frequency studies were too coarse to detect at all.

Mark Allocation — Part (b) (50 marks)

Tier	Marks	What separates this tier
38-50	38-50	Correctly distinguishes volatility timing from return timing AND correctly describes the Busse (1999) model and why daily (not monthly) data is required AND correctly interprets the model's key coefficient AND correctly states the empirical findings (successful volatility timing, higher Sharpe ratios, stronger among surviving funds).
25-37	25-37	Correctly distinguishes volatility timing from return timing with a broadly correct model description, but the empirical findings or the daily-vs-monthly data point is missing.
13-24	13-24	Vague description of "managers adjusting risk over time" without a specific model or findings.
0-12	0-12	No credible model of volatility timing given.

Question A2 Total: 100%

Section A · Question 3

Testing for performance persistence — 100 marks

A3(a)

[25 marks]

Distinguish between statistical and economic significance in tests of portfolio performance persistence.

A3(b)

[50 marks]

Describe any two tests of persistence. Explain in your answer whether these are tests of statistical or economic significance or both.

A3(c)

[25 marks]

A finding of persistence provides an easy 'beat the market' strategy for investors. Do you agree with this statement? Explain your answer.

FULL MODEL ANSWER

EXAM TECHNIQUE

This entire question wasn't set out in the uploaded course material; the discussion below follows the standard mutual fund persistence literature (Hendricks, Patel and Zweig, 1993; Malkiel, 1995; Brown and Goetzmann, 1995; Carhart, 1997).

Statistical significance: asks whether an observed pattern of persistence — past winning funds continuing to win, or past losing funds continuing to lose — is large enough, relative to its sampling variability, to be confident it did not arise by pure chance. This is assessed with standard hypothesis-testing tools: t-statistics on a persistence coefficient, chi-square tests on a winner/loser contingency table, or confidence intervals on a winner-minus-loser return spread.

Economic significance: asks a genuinely different question: even if a persistence pattern is statistically real (not chance), is it **large enough in magnitude** to actually matter to an investor in practice — specifically, is the winner-minus-loser return spread large enough to survive real-world **transaction costs, fund fees/expense ratios, loads, and taxes**, and still leave a genuinely exploitable excess return?

Why the distinction matters — these can pull in opposite directions: with a large enough dataset (many funds, many years), even a very small persistence effect can be estimated precisely enough to be **statistically** significant, while still being far too small, once realistic costs are subtracted, to be **economically** significant — meaningless as an actual investment strategy. The reverse is also possible: a persistence effect estimated from a small sample might be economically large but fail to reach conventional statistical significance simply because the sample doesn't provide enough statistical power to distinguish it confidently from noise. A rigorous test of persistence needs to establish **both**, since neither alone answers the practical question an investor actually cares about.

Mark Allocation — Part (a) (25 marks)

Tier	Marks	What separates this tier
19-25	19-25	Correctly defines statistical significance (confidence the pattern isn't chance) AND correctly defines economic significance (magnitude large enough to survive real-world costs) AND explains why a result can be one without being the other.

13-18	13-18	Both concepts correctly defined, but the relationship/tension between them is not explained.
7-12	7-12	Only one of the two concepts correctly defined.
0-6	0-6	No credible distinction given.

Test 1 — the contingency table / cross-product ratio approach

(Malkiel, 1995; Brown and Goetzmann, 1995): funds are classified as "winners" (W) or "losers" (L) in each of two consecutive periods, based on whether their return (or risk-adjusted alpha) is above or below the median. This produces a 2x2 contingency table (WW, WL, LW, LL counts). If there is no persistence, a fund's classification in period 2 should be **independent** of its classification in period 1, so the table should show no more WW/LL combinations than chance alone predicts — tested formally with a **chi-square test of independence**, or equivalently a cross-product ratio (odds ratio) test on the table's diagonal versus off-diagonal cells. **This is purely a test of statistical significance** — it tells you whether the classification pattern is statistically different from random, but says nothing directly about the size of any resulting return advantage.

Test 2 — the recursive/rolling portfolio approach — "hot hands"

(Hendricks, Patel and Zweig, 1993): funds are ranked into deciles by their performance over a past ranking period, and a portfolio of the top-decile ("hot hand") funds is tracked forward over a subsequent evaluation period, compared against a portfolio of bottom-decile funds — the process is then rolled forward repeatedly through the sample. This test is **both statistical and economic**: the *statistical* question is whether the average return spread between the top- and bottom-decile portfolios is significantly different from zero (a standard t-test on the spread); the *economic* question — which this method directly and naturally speaks to, unlike the contingency-table approach — is the actual **size** of that spread in percentage-point terms, which can be compared directly against realistic transaction costs and fund fees to see whether it represents a genuinely exploitable strategy.

Mark Allocation — Part (b) (50 marks)

Tier	Marks	What separates this tier
38-50	38-50	Two distinct persistence tests correctly described in enough detail to show how each is actually implemented AND each correctly classified as testing statistical significance, economic significance, or both, with the reasoning for that classification given.
25-37	25-37	Two tests correctly described, but the statistical/economic classification is asserted rather than reasoned through.
13-24	13-24	Only one test correctly described in adequate detail.
0-12	0-12	No credible tests of persistence described.

Largely disagree — persistence, even where statistically genuine, does not straightforwardly translate into an easy, exploitable beat-the-market strategy net of real-world costs and its own well-documented sources.

The costs problem — this is exactly the statistical vs. economic significance distinction from Part (a):

even a statistically robust persistence finding may not survive the transaction costs of actually implementing a winner-chasing strategy (buying and frequently rebalancing into recent top performers), or the expense ratios and loads charged by the winning funds themselves — a persistence pattern documented in gross, pre-

cost returns can shrink substantially, or disappear entirely, once realistic implementation costs for an ordinary investor are subtracted.

The Carhart (1997) critique — persistence often isn't skill at

all: Carhart's landmark study found that most of the apparent persistence in mutual fund performance is explained not by genuine, repeatable stock-picking skill, but by (i) persistent differences in **expense ratios and transaction costs** across funds (a fund with structurally lower costs will mechanically show up as a "winner" repeatedly, for reasons that have nothing to do with skill), and (ii) funds' incidental, ongoing exposure to the **momentum factor** (see Q. A1(b)) — a fund that happens to be tilted toward recent winning stocks will keep looking like a persistent winner for as long as the momentum effect itself persists, which is a factor exposure, not genuine selection skill. Once these are controlled for, Carhart found **little evidence of persistent risk-adjusted outperformance** beyond the first year — and found that persistence is often considerably **stronger among persistent losers** (bad funds reliably stay bad) than among persistent winners, which is not a strategy an investor can profitably exploit on the winning side at all.

A further, structural caveat — persistence is a backward-looking, sample-specific finding: even a genuine, cost-surviving persistence pattern documented *within a historical sample* is not guaranteed to continue *out-of-sample* — fund flows chasing yesterday's winners can themselves erode a strategy's edge (too much new capital pouring into a successful, capacity-constrained strategy), and any strategy popularised in response to a documented anomaly risks being arbitrated away exactly as the module's own EMH material discusses for other anomalies.

Mark Allocation — Part (c) (25 marks)

Tier	Marks	What separates this tier
19-25	19-25	Correctly disagrees (or gives a well-reasoned partial agreement) AND engages the costs/economic-significance problem specifically AND cites the Carhart critique (expense ratios and momentum exposure explaining apparent persistence, rather than genuine skill) AND notes persistence is often stronger among losers than winners.
13-18	13-18	Correctly disagrees with at least one substantive reason (costs or the Carhart critique) given, but not both.
7-12	7-12	Simply agrees with the statement, or disagrees without any substantive reasoning.
0-6	0-6	No credible answer given.

Question A3 Total: 100%